

Group030 AI Records Index

FIT5196 Data Wrangling | Assessment 1 | Semester 2, 2026

Conversational-AI record: AI-01 (first assignment-related ChatGPT conversation)

Important submission note

This index record does not replace the complete conversation export. For submission, place this AI index (converted to PDF) together with the complete, unshortened ChatGPT conversation export in the AI_records/ folder. Do not use selected prompts, a summary, or screenshots as a substitute for the complete export.

Conversational-AI index

Record ID	Member, tool/model and complete export filename	Purpose and affected submission file/section	Independent verification
A1-01	Member: Xinhang Ren 35134410 Tool: ChatGPT (OpenAI) Model: GPT-5.6 Sol Complete export filename: A1-01.pdf	Purpose: Review the official FIT5196 A1 Specification and Marking Rubric and develop a step-by-step, HD-oriented completion plan. The conversation covered: structured JSON/XML parsing and source profiling; source-to-target mapping; the six required relational outputs; duplicate/cross-source reconciliation; prescribed arithmetic; regex and multilingual Unicode processing; executable validation; reproducibility; 6-8 assessed EDA visualisations; exactly 10 evidence-based findings; exactly 5 ML questions; and final submission structure. Affected work: planning and quality-control guidance for Group030_solution.ipynb, Group030_source_to_target_mapping.csv, Group030_text_functions.py, Group030_EDA.ipynb, Group030_EDA.pdf, the validation register, and the final submission checklist.	1. Requirements and thresholds were checked against the official FIT5196 A1 Specification and Marking Rubric.
			2. The implemented text-processing functions were checked using public cases TXT-01-TXT-18 (18/18 passed) and additional student edge/near-match/Unicode tests.
			3. The transformation outputs were checked using the VAL-* validation register (56/56 PASS), including schema, PK/FK, overlap/conflict, arithmetic, temporal, text and multilingual checks.
			4. Relational EDA joins were checked with join/cardinality assertions to guard against row multiplication.
			5. Both notebooks were independently re-run offline using a fresh Restart & Run All / clean-directory workflow to confirm reproducibility.
A1-02	Member: Sizhe Hong 35381949 Tool: ChatGPT (OpenAI) Model: GPT-5.6 Sol Complete export filename: A1-02.pdf	Purpose: Review the Group030 source data and develop a complete validation framework tailored to the actual JSON/XML integration workflow. The conversation covered: schema conformity; data type and format checks; nullable/non-nullable fields; PK uniqueness; FK and referential integrity; source-level duplicate detection; raw-to-deduplicated-to-overlap-to-final row-flow reconciliation; cross-source JSON/XML overlap resolution; arithmetic consistency; temporal and domain checks; categorical constraints; regex extraction; multilingual Unicode/text-cleaning validation; and	1. Schema recommendations were checked against the published data dictionary, including the required six output tables, column structure, grain and nullable constraints.
			2. Source row counts and duplicate/overlap calculations were independently reconciled using stable identifiers rather than assuming that the JSON and XML sources could simply be concatenated. For orders, both sources contained 2,750 unique order IDs with 500 shared IDs, giving 5,000 canonical orders.

		advanced semantic checks such as warehouse-distance consistency. The analysis also established expected canonical row counts and reconciliation targets for the six required relational outputs, including 5,000 orders, 15,723 order items, 500 customers, 5,000 deliveries, 1,000 products and 7,000 product reviews. Affected work: validation and quality-control sections of Group030_solution.ipynb; validation assertions for the six required output tables; source reconciliation/row-flow evidence; PK/FK integrity checks; text-processing validation; and validation evidence used in the final report/EDA discussion.	3. Equivalent reconciliation was checked for reviews and order items: 3,850 unique reviews per source with 700 overlapping IDs produced 7,000 canonical reviews; item-level unique counts and 1,557 shared items produced 15,723 canonical order items. 4. PK/FK integrity was checked across the relational model, including orders→customers, order_items→orders/products, deliveries→orders, and reviews→orders/order_items/products/customers; semantic cross-key consistency was also recommended to prevent valid-but-mismatched foreign-key combinations. 5. Arithmetic and semantic checks were independently defined for line revenue, order price, order total, delivery timing, review timing, ranges, regex extraction and text cleaning. Final validation was designed to report explicit PASS/FAIL results and violation counts rather than relying on visual inspection alone.
A1-03	Member: Yinglin Fang 34814248 Tool: ChatGPT (OpenAI) Model: GPT-5.6 Sol Complete export filename: AI-03.pdf	Purpose: Review the Group030 source-data structure and determine the correct relational decomposition for the cleaned dataset, with particular emphasis on table grain, primary/foreign keys, field allocation and downstream validation requirements. The conversation established the six required canonical cleaned tables: orders, order_items, customers, deliveries, products, and product_reviews, and clarified the intended grain and PK/FK relationships for each table. It also covered field-level cleaning requirements, numeric and date standardisation, boolean conversion, multilingual text cleaning, regex extraction, arithmetic reconciliation, temporal consistency, and referential-integrity checks. The discussion further clarified that JSON and XML should be treated as source systems rather than separate final outputs, and that both sources should be normalised and reconciled into one canonical six-table model before validation. Affected work: relational table design and cleaning logic in Group030_solution.ipynb; schema and field mappings in Group030_source_to_target_mapping.csv; cleaning rules in Group030_text_functions.py; and the validation sections covering schema, PK/FK, arithmetic, temporal, text/regex, source-row-flow and overlap reconciliation.	1. The proposed six-table structure was checked against the published Group030 data dictionary and required cleaned-output structure, including table grain, primary keys, foreign keys and expected fields 2. PK/FK relationships were independently checked using uniqueness and referential-integrity assertions, including orders.customer_id → customers.customer_id, order_items.order_id → orders.order_id, order_items.product_id → products.product_id, and review-related references. 3. Arithmetic relationships were verified independently, including line_revenue ≈ quantity × unit_price and reconciliation between aggregated order-item revenue and the corresponding order-level price. 4. Temporal rules were checked independently across order and delivery fields, including logical ordering of order, dispatch, promised and delivered dates and consistency of derived delay measures. 5. The final pipeline structure was independently reviewed to ensure that JSON/XML source records are normalised, deduplicated and reconciled before being written to the six canonical tables, with validation covering schema, types, missing values, PK/FK integrity, source/row flow, overlap, arithmetic, temporal, text/regex and domain/range checks.

A1-04	Member: Guohou Zhang 35800275 Tool: ChatGPT (OpenAI) Model: GPT-5.6 Sol Complete export filename: A1-04.pdf	Purpose: Review the Group030 source data and obtain dataset-specific recommendations for Regex validation and multilingual Unicode/text cleaning. The conversation covered identifier-format validation for customer, product, order, item, delivery, review and supplier IDs; SKU and source-record formats; coupon codes; Australian postcodes; language codes; JSON/XML dates and timestamps; AUD monetary values; percentage fields; URLs, HTML tags and embedded review metadata. It also covered Unicode NFKC normalisation, HTML entity decoding, whitespace/NBSP handling, invisible/control-character detection, emoji handling, and preservation of multilingual text rather than destructive ASCII stripping. Affected work: validation and text-cleaning logic in Group030_text_functions.py, relevant validation-register entries, and the Regex/Unicode cleaning and validation sections of the submission.	1. Regex recommendations were checked against representative values from the Group030 source data and the expected field formats, including IDs, SKU, dates, monetary values, discounts and language codes. 2. Regex-based syntactic checks were separated from semantic/domain validation; for example, date strings were additionally parsed as dates, postcodes were range/domain checked, and language codes were checked against an allowed set rather than relying on pattern matching alone. 3. Unicode recommendations were checked to ensure multilingual text was preserved, including Chinese, Japanese, Arabic, Russian, Hindi and accented Latin text, instead of applying destructive ASCII-only conversion. 4. Text-cleaning steps were independently reviewed for NFKC normalisation, HTML entity decoding, HTML/URL removal, whitespace normalisation, and detection of invisible/control characters. 5. Raw-versus-clean-field handling was checked so that cleaning operations were applied to derived/canonical text fields while preserving the original source text for traceability and reproducibility.
A1-05	Member: King Man Chan 36550779 Tool: ChatGPT (OpenAI) Model: GPT-5.6 Sol Complete export filename: A1-05.pdf	Purpose: Review the provided public text-processing test cases and develop additional student-designed validation cases for Group030's text/regex processing functions. The conversation focused on identifying gaps not already covered by the public cases and designing complementary matched, unmatched, near-match, missing-value, multilingual and Unicode edge cases. It covered narrative-text cleaning, recognised system-marker removal, URL/HTML/emoji handling, order-reference extraction, product-SKU extraction, promotional-code extraction, non-Latin-script detection and Latin-analysis text construction. Boundary-sensitive regular-expression patterns and negative cases were also considered to prevent partial or over-broad matches. A recommended subset of student-designed tests was selected to maximise coverage while minimising duplication with the supplied public tests. The conversation also proposed a reproducible test-result structure containing case ID, function, input, expected output, actual output, PASS/FAIL status and test purpose.	1. Student-designed tests were compared against the supplied public test suite to ensure that they added complementary coverage rather than simply reproducing existing cases. 2. The proposed cases were independently checked against the published text-processing requirements, including valid and invalid order references, product SKUs and promotional codes, as well as missing, multilingual and Unicode inputs. 3. Regex boundary behaviour was verified using near-match and negative examples, including embedded order references, over-length identifiers, invalid promotional-code ranges and malformed SKU strings, to ensure that invalid strings could not produce partial valid matches. 4. Narrative-cleaning tests were checked to ensure that only recognised markers were removed and that arbitrary bracketed customer wording was retained, preventing the use of an overly broad bracket-removal regex.

		Affected work: student-designed validation cases and related quality-control evidence in Group030_solution.ipynb / text-processing validation section, with corresponding checks for Group030_text_functions.py and the validation evidence reported in the final submission.	5. The final student-designed cases were intended to be executed through the same validation workflow as the public tests, with expected versus actual outputs and explicit PASS/FAIL evidence recorded in the notebook.
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Before submission

- Replace the member placeholder with the actual full name and student ID.
- Replace the model placeholder with the exact model name shown in the original ChatGPT export, if the interface records it.
- Export the complete first assignment-related conversation without shortening it and use the exact export filename shown above, or update this index to match the actual filename.
- If other assignment-related AI conversations exist, add AI-02, AI-03, etc.; one complete conversation may support several purposes or submission locations.
- Convert the final English AI index to Group030_AI_index.pdf and place it in AI_records/ together with every complete conversation export.
- Complete and sign Group030_AI_declaration.pdf separately.

Suggested AI_records/ folder

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AI_records/
  Group030_AI_index.pdf
  A1-01.pdf
  A1-02.pdf
  A1-03.pdf
  A1-04.pdf
  A1-05.pdf
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